

BUILD INSTRUCTIONS · V3

Contacts, file automation and trace notes

For QuantumView_Checkpoint_DateSlicersPassed.pbix. Nine steps, about 50 minutes. Everything here is model-layer, so there is no new .pbix — keep working in your own file.

No new .pbix this round, on purpose

Your file already opens and still carries its sensitivity label. Handing back a label-stripped copy would cost you the marking and gain you nothing — every fix below is a query or a measure. The one report-layer thing that is broken fixes itself: two date slicers are bound to 'Dim Date'[Date] and that table does not exist. Step 5 creates it and they light up.

What you got right, so I did not touch it

The roster build is the hard part of this and it is correct. Branch_Contacts_All to Branch_Contacts to Trace_Contact_By_FA is the right shape, and critically the BOA-to-FA join is on the right key:

Join	Resolves
BOA[FA ID] → FA[EMPLID]	24,972 of 25,007 · 99.9%
BOA[FA Name] → FA[Contact Name]	4,930 of 25,007 · 19.7%

Name-matching was the obvious approach and it would have quietly lost four fifths of the pairings.

Trace_Contact_By_FA is untouched by every script in this pack.

At a glance

What is wrong	Impact	Fixed by
Trace FA numbers never match the roster — 0 of 95	Every contact measure empty	Step 3
Trace_Notes reads an abandoned workbook	199 blank rows, 1 of 95 join	Step 9
QV_RAW points at a different QV_Data than your path sheet	Possibly reading a dead folder	Step 2
QV_RAW would read every export once per subfolder	Total Shipments multiplies	Step 2
Morning/afternoon defined twice, at 17:00 and 13:30	Two answers, same rows	Step 2
Two account relationships silently deactivated	Vendor slicer filters nothing	Step 1
Dim Date does not exist but two slicers bind to it	Two dead slicers	Step 5
QV_Manifest dedupe still unbuffered	Oldest row per shipment wins	Step 2

SECTION 1

The one thing that is broken

Current_Open_Trace[FA #]	I287748
Trace_Contact_By_FA[FA #]	287748

The relationship between them is **active and matches 0 of 95 rows**. That single character is why Selected Branch Email, Selected FA Email, Selected BOA Emails and Unreachable Traces all come back empty, and why the Unreachable card on ACTIVE TRACES reads 95 — everything is unreachable.

Strip the prefix and **66 of 66** FA-numbered traces match, resolving an FA address on all 66 and a BOA cc on 61.

The prefix is a type tag

One column is carrying two identifier spaces and one data-entry miss:

Prefix	What it is	Traces	Routes to
I#####	a real FA number	66	the roster
H#####	a home-office department code	28	its own map — the roster has none of them
0	a data-entry miss	1	nowhere, and it should say so

The 28 home-office rows are not FAs

Tested against every candidate column in the roster:

Against	Matches
EMPLID	0 of 28
JP Number	0 of 28
FA ID	0 of 28
FA #	0 of 28
Branch Number	5 of 28 — coincidence, not a route

All 28 are Location = Campus Ship / World Ship / Kimler — home-office departmental shipments. Only **11 distinct codes**, and **seven already have a confirmed address** sitting in CONTACT INFO on their own trace rows. Those are seeded in Home_Office_Contacts; four need somebody to fill in an address.

What it is worth

	Now	After
Traces with a resolvable address	0 via roster	90 of 95
— via FA roster	0	66
— via home-office map	0	24
With a BOA cc	0	61 of 66
Still blocked	95	5, each naming its own reason

Contact Route is a new column that makes the branch explicit rather than leaving it implied in a chain of fallbacks, so a trace that cannot be addressed says *why* instead of quietly not appearing in the queue.

SECTION 2

Your paths, reconciled

Three of the locations on your sheet disagree with what the queries actually read. Full detail in PATHS.md.

1 · QV_RAW is pointed at a different folder

Path	
Your path sheet	...\Administrative Services-Solutions\Power BI Reporting\Report Data\QV_Data
QV_RAW in the .pbix	...\Administrative Services-Solutions\Quantum View\QV_Data

Different parent folder. Check which one your last successful refresh actually read — **if the old path still resolves, you have two QV_Data folders and only one of them is being fed.**

2 · It must point at Archive, not QV_Data

Folder. Files **recurses**, and QV_Data already contains Morning QV, Afternoon QV, Processed and Failed. Pointed at QV_Data the query reads every export once per folder it appears in — Total Shipments multiplies and nothing errors. It would also try to parse QV_Automation_Helper_Lists.xlsx as a 34-column CSV.

This is not hypothetical. Those folders exist on your share today, which is what turns Archive-only from a precaution into a requirement.

3 · The notes query is reading a dead file

Trace_Notes reads ... \Report Data\QV_Tracing_API_Aligned_Workflow_Optimized.xlsx
The notes live in Trace_Master_Streamlined_ (SharePoint)

That is the entire explanation for 199 blank rows out of 200 and a join that lands on 1 of 95 traces. Nothing was broken — the query was pointed at the abandoned copy. 199 blank rows is what an abandoned sheet looks like once somebody formats past the data.

Move the trace list off the personal OneDrive before rollout

https://ejprod-my.sharepoint.com/personal/p258239_edwardjones_com/...

That is one individual's OneDrive, not a team site. It is **deprovisioned when that person leaves or changes roles** — read-only, then gone, typically after a 30 to 93 day retention window, and the notes history goes with it. Service refresh also has to run on **that person's** credentials: it cannot be moved to a service account, because the site belongs to the human. Every other "use a service account" note in this build is a policy choice; this one is a structural block.

The query works against it today and I am not holding the build up over it. But copy the list to a team site and change SiteUrl — ten minutes, and every column name survives the move.

Rosters — left alone deliberately

Your sheet gives X:\hrdi_report_pickup; Branch_Contacts_All reads Report Data\Rosters. The second one **works** — 46,080 rows across 21,073 FAs and 25,007 BOAs, refreshed 06:49 the morning you sent the file. Breaking a working roster to remove a copy step is a bad trade this close to rollout, so it is documented rather than changed.

Watch [Latest Roster Modified]. If it stops moving, the copy from the pickup folder stopped.

Two constants now live in two files

The **13:30 cutoff** and the **QV_Data root** are hard-coded in both qv-file-automation.ps1 and section 1 of model-fixes-v3.pq. Change one and you must change the other, or the share and the dashboard will disagree about which run a file belongs to — and nothing will error.

Use UNC, not X:. Drive letters are mapped per logon session. The Power BI Service has none, and a scheduled task running as the service account with nobody logged on has none either — it fails with a path-not-found that looks like a permissions problem and is not.

SECTION 3

The runbook

Nine steps, about 50 minutes. Do them in order — each gates the next. Keep a copy of your .pbix before you start.

Copy code from the script files, not from this PDF

A PDF renders straight quotes as curly quotes and hyphens as en-dashes. M and DAX both reject those, and the error message will not tell you that is what happened. Open the .pq, .tmdl and .ps1 files from quantum-view-v3-scripts.zip in Notepad++, VS Code or even Notepad.

Step 1 · Model view — four edits

3 min

TMDL cannot script these safely. Power BI allows only one **active** relationship between a given pair of tables, so scripting a new one while the old is live either errors or deactivates something you wanted.

a. Delete Current_Open_Trace[FA #] → Trace_Contact_By_FA[FA #]. It matches 0 of 95. Step 3 replaces it with [FA Key].

b. Set inactive the two Exception Key → Resolutions Table relationships, then **set active** the two Account Key → Dim_Account ones.

Both account relationships are inactive right now, which is why the **Vendor Acct # slicer on ACTIVE TRACES filters nothing**. Power BI deactivated them when Trace to Resolutions created a second path to Dim_Account. Measured on your data:

Path to Dim_Account	Covers
Account Key → Dim_Account (direct)	95 of 95
Exception Key → Resolutions → Dim_Account	60 of 95

The direct one wins. The Exception Key link stays in the model, inactive, for USERRELATIONSHIP when you genuinely want "traces for the exceptions I selected".

c. Untick Enable load on Branch_Contacts_All — 46,080 staging rows, referenced by no visual, and the third side of a triangle holding Trace_Contact_By_FA[Branch Number] inactive. **d. Set active** that relationship once (c) removes the ambiguity.

✓ Model view shows no dashed relationship you did not intend.

Step 2 · Power Query — the file side

10 min

Home > Transform data, then **scripts/model-fixes-v3.pq**.

§	Query	What changes
1	QV_RAW (<i>replace</i>)	Corrected path, reads Archive only, computes the run window once at 13:30.
2	QV_Output (<i>two steps</i>)	Delete #"QV Run Time" and #"QV Run Date" — they now come from §1. Rename Run Window to Run Time so no visual needs rebinding.
3	QV_Manifest (<i>two steps</i>)	Table.Buffer on the dedupe — it is still keeping the oldest row per shipment; the fix landed on Resolutions Table but not here. And move Account Key from DAX to M.

✓ **Total Shipments is unchanged** after the refresh. If it jumped, the Archive path is wrong.

Step 3 · Power Query — the contact fix

10 min

§	Query	What changes
4	Current_Open_Trace (<i>add at end</i>)	Three columns: FA Key, Home Office Code, Contact Route.
5	Trace_Active (<i>add at end</i>)	The identical block, so the two trace tables stay interchangeable. If only one can resolve a contact they will disagree the first time somebody builds on the wrong one.
6	Home_Office_Contacts (<i>new</i>)	Eleven codes; seven already answered by addresses in your own CONTACT INFO column. Reads a sheet from QV_Automation_Helper_Lists.xlsx if one exists, falls back to the inline seed if not — so it works unedited today and moves to the workbook without anyone touching M.

✓ Contact Route shows four values — Case contact, FA roster, Home office, Unroutable — with **Unroutable at exactly 1**.

Step 4 · Power Query — the send queue

5 min

§8 Mail_Merge_Queue — new. One row per trace, already addressed. Send To and Send Cc are semicolon strings, which is exactly what the Outlook connector's To and Cc fields take — no splitting, no lookups in the flow.

Every join, fallback and exclusion decision lives in M where you can read it, rather than scattered across flow actions where you cannot. **Blocked Reason** is the point of the table: a trace that cannot be emailed says why, in a column.

Needs §4 and §6 in place first.

✓ 95 rows — 90 with a Send To, 5 with a Blocked Reason.

Step 5 · Apply the model script

5 min

View > TMDL view → paste **scripts/model-updates-v3.tmdl** → read it → **Apply**.

Contact measures rewritten onto the working keys — **same names, so nothing needs rebinding** — plus the coverage measures, the mail-merge measures, Dim Date and its four date roles.

- ✓ Dim Date appears and starts in 2026. **The two date slicers work.**
- ✓ Mark it: Dim Date > Table tools > Mark as date table > Date.

Step 6 · Verify

5 min

View > DAX query view. **checks/10-contact-routing.dax** is the one that matters.

	Now	Should read
Total Active Traces	95	95
Contactable Traces	0 via roster	90
Unreachable Traces	95	5
BOA Coverage	0	61
Sendable + Blocked	—	90 + 5

The fourth query in that file is the blocked worklist — four home-office codes with no address, plus the one trace whose FA # is the literal string "0". It is short, actionable, and every row names its own fix.

Step 7 · Clean up

2 min

Untick Enable load, or delete	Why
Branch_Contacts_Old	500 rows, every column null on every row
Exception Reason Map	loaded and the source of Dim_ExceptionReason — the same 14 rows twice in the Fields pane, carrying two relationships that exist only as inactive
Res_Tracker	1,040 rows of unpromoted Column1 to Column14

And add **A25T52** to ...\\Report Data\\UPS Acct Info .xlsx, extending the table range to A1:L25. It is still the entire unmatched remainder — 755 manifest rows and 102 queue rows.

Step 8 · The file automation

15 min, once

scripts/qv-file-automation.ps1, using the folders you already have:

```
Report Data\\QV_Data\\
  Archive\\                <- exports land here. THE MODEL READS THIS.
  Morning QV\\             <- copies, before 13:30
  Afternoon QV\\           <- copies, at or after
  Processed\\_processed.csv <- state, so re-runs are cheap
  Failed\\                 <- copies of problems, plus a .reason.txt
  _logs\\                  <- one log per month
  QV_Automation_Helper_Lists.xlsx <- not touched by the script
```


It **copies, never moves**, and never writes to Archive at all. Archive stays the source of truth, so if the script fails, is disabled, or double-fires, the dashboard is unaffected. That is the whole design — and it is why even a corrupt file gets *copied* to Failed rather than moved out of Archive.

1. `.\qv-file-automation.ps1 -WhatIf` — see what it would do
2. Run it for real, check `_logs\`
3. Schedule it — Task Scheduler, 09:00 and 15:00, repeating every 30 minutes for 2 hours so a late export gets picked up. Full instructions in the script footer.

Use the service account, not your login — same rule as the gateway credentials. And **UNC, not X:**.

Step 9 · Notes, then the flows

5 min + the flows

\$7 repoints `Trace_Notes` at `Trace_Master_Streamlined_`. Paste the first two steps, load, and look at the column names SharePoint gives you — they are **internal** names, so "Trace Status" arrives as `Trace_x0020_Status`. Correct the renames, then paste the rest.

Then replace the single `CASE #` merge in `Current_Open_Trace` with two passes: Exception Key first, `CASE #` for whatever is left, **both sides normalised**. One key is stable and the other is the one analysts actually type — you need both, or you get today's 1 of 95.

`TRACE-NOTES-AND-MAILMERGE.md` has the list schema, both flow definitions, the templates and the blocked-tenant fallback. **Check first whether your tenant permits Flows** — it decides whether that part is two days or two weeks.

SECTION 4

Still open after this

	Why it is still open
Trace date columns are still text	model-fixes-v2.pq §10. Every date column on both trace tables loads as text, so no calendar can reach the trace side — no date slicer, no trend, no SLA maths on the page whose entire job is SLA. This is the largest remaining structural gap.
Four Home_Office_Contacts addresses	H03130, H05057, H05150, H44804. Filling them takes reachability from 90 to 94 of 95.
One trace with FA # = "0"	Data entry on the trace sheet. It can never resolve to a contact.
Move the trace list to a team site	Personal OneDrive cannot be refreshed by a service account and is deprovisioned when its owner leaves.
Root Cause / Next Action / Notes 0% populated	The exceptions-side write-back — a different list from the tracing notes. WRITEBACK-AND-MAILMERGE.md.
Exception grain decision	8,583 distinct keys vs 29,957 rows flagged Is Issue vs 45,614 total rows. Needs a business answer, not a build.
A25T52 in the account workbook	The entire unmatched remainder on both fact tables.

The document pack

File	What it is
V3-RUNBOOK.md	this document, in markdown
PATHS.md	every location, and the three that disagree with the model
QUERY-VALIDATION.md	all 14 queries and every relationship, measured
TRACE-NOTES-AND-MAILMERGE.md	the notes list, both flows, FA/BOA addressing rules
WRITEBACK-AND-MAILMERGE.md	the exceptions-side write-back. A different list — do not merge the two
PBIT-GUIDE.md	using a .pbit template with a .pbix
CURRENT.md	the entry point

The v2 pack is superseded

UPS_QuantumView_Build_Instructions.pdf and NEW-BUILD-FIXES.md describe the previous file — NEWQuantum_View_Exceptions_Dashboard_fixed.pbix — and a 12-step runbook whose model work you have already applied. Keep them for the reasoning behind the classifier, the dedupe direction and the account key derivation. **Do not follow them as instructions, and do not re-run model-fixes-v2.pq sections 1 to 9 or model-updates-v2.tmdl** — the only part of v2 still outstanding is section 10, the trace date columns.